

Constructing the Future in and of Cyberbiosecurity: Cyber-Bio Convergence and (In)Security Imaginaries

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Perceptions of cyberattacks leading to catastrophes and disasters have been integral to processes of securitisation and militarisation of cybersecurity for many years. Recent digitisation processes in life sciences (e.g., biology, biotechnology, pharmaceutical and biomedical domains, among others) have extended such perceptions to the vulnerabilities and dependencies generated by this “cyber-bio” convergence, that may spread diseases, enable the development of biological weapons, or cause deaths. Accordingly, in recent years, “cyberbiosecurity” has emerged as a new field of research and policy analysis to study such cyber-bio risks and develop ways to prevent them through pre-emptive security measures. This article interrogates the construction of the “future” *in* and *of* cyberbiosecurity as presented in scholarly literature in the life sciences. It examines the sociotechnical imaginaries of cyber-bio convergence, on one hand, and the (in)security imaginaries of future cyber-bio risks, on the other, as two competing yet interdependent narratives that are mutually co-constitutive. In doing so, the article critiques cyberbiosecurity imaginaries for their limited engagement with ontological and epistemological questions posed by cyber-bio infrastructures as complex *informational systems*. It argues that cyber-bio convergence introduces an *informationalist challenge* that reconfigures core assumptions about actancy, referent objects, and uncertainty, in security analysis, with potential transformative implications for cybersecurity scholarship and governance.

Las percepciones de que los ciberataques pueden causar catástrofes y desastres han sido parte fundamental de los procesos de securitización y militarización de la ciberseguridad durante muchos años. Los procesos recientes de digitalización que han tenido lugar en las ciencias de la vida (por ejemplo, biología, biotecnología, farmacéutica y ámbitos biomédicos, entre otros) han ampliado estas percepciones hacia las vulnerabilidades y dependencias generadas por esta convergencia «ciber/bio», con respecto a que podrían propagar enfermedades, permitir el desarrollo de armas biológicas o causar muertes. En consecuencia, en los últimos años, la «ciberbioseguridad» ha surgido como un nuevo campo de investigación y de análisis de políticas que tiene el objetivo de estudiar estos riesgos ciber/bio y desarrollar formas de prevenirlos mediante medidas de seguridad preventivas. Este artículo analiza la construcción del «futuro» *de* la ciberbioseguridad y también *en* la ciberbioseguridad, tal como se presenta en la literatura académica de las ciencias de la vida. El artículo analiza los imaginarios sociotécnicos de la convergencia ciber/bio, por un lado, y los imaginarios en materia de (in)seguridad de los riesgos ciber/bio futuros, por otro, como dos narrativas que compiten entre sí, pero que son interdependientes y se coconstituyen mutuamente. Con ello, el artículo critica los imaginarios de la ciberbioseguridad por su limitado nivel de compromiso con las preguntas ontológicas y epistemológicas que plantean las infraestructuras ciber/bio como *sistemas informacionales* complejos. El artículo argumenta que la convergencia ciber/bio introduce un desafío *informacionista* que reconfigura las suposiciones básicas sobre la agencia, los objetos de referencia y la incertidumbre en el análisis de seguridad, lo que tiene posibles implicaciones transformadoras para la investigación académica y la gobernanza en materia de ciberseguridad.

Le fait que l'on pense que les cyberattaques peuvent conduire à des catastrophes ou désastres fait partie intégrante des processus de sécuritisation et de militarisation de la cybersécurité depuis de nombreuses années. À cause de récents processus de numérisation en sciences de la vie (par exemple, en biologie, en biotechnologie, et, entre autres, dans les domaines pharmaceutique et biomédical), ces perceptions s'appliquent désormais aux vulnérabilités et dépendances générées par la convergence « cyber-bio », qui pourrait transmettre des maladies, permettre le développement d'armes biologiques ou engendrer des morts. Par conséquent, ces dernières années, la « cyberbiosécurité » se dessine comme un nouveau domaine de recherche et d'analyse politique pour étudier ces risques cyber-bio et trouver des moyens de les prévenir par des mesures de sécurité préventives. Cet article interroge la construction de l'« avenir » *dans* et *de* la cyberbiosécurité telle qu'elle est présentée dans la littérature académique en sciences de la vie. Il examine les imaginaires sociotechniques de la convergence cyber-bio d'une part, et les imaginaires de sécurité et d'insécurité des futurs risques cyber-bio de l'autre, comme des récits concurrents mais interdépendants, mutuellement co-constitutifs. Ce faisant, l'article critique les imaginaires de cyberbiosécurité sur leur traitement limité des questions ontologiques et épistémologiques posées par les infrastructures cyber-bio en tant que *systèmes informationnels* complexes. Il affirme que la convergence cyber-bio introduit un *défi informationnaliste* qui reconfigure des hypothèses fondamentales concernant l'actance, les référents et l'incertitude en analyse de sécurité, ce qui s'accompagne potentiellement d'implications transformatives pour la recherche et la gouvernance relatives à la cybersécurité.

Keywords: cybersecurity; biosecurity; cyber threats; biological dangers; global health; sociotechnical imaginaries.

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Fouad, Noran Shafik (2026) Constructing the Future in and of Cyberbiosecurity: Cyber-Bio Convergence and (In)Security Imaginaries. *Global Studies Quarterly*,

<https://doi.org/10.1093/isagsq/ksag101>

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Introduction

The future of cyberbiosecurity is very bright (Murch 2023, 14)

In 2023, 23andMe, one of the biggest companies in direct-to-consumer genetic testing (DTC-GT), was the target of a major data breach.¹ At first, it appeared that the breach only affected 14,000 customers who sent their DNA samples to be analyzed by the company for ancestry and health insights. However, because many customers' accounts are linked to their relatives, 23andMe revealed later that the actual number of those affected is 6.9 million. The stolen data included names, birthdates, ancestry reports, relationship labels, among others (Franceschi-Bicchierai 2023). Moreover, some of the stolen data was released on a cybercrime forum and sold for \$1–10 per account. Among the leaked data, one million data points were specifically about Ashkenazi Jews, and hundreds of thousands others were linked to users of Chinese heritage (Newman 2023). In March 2025, 23andMe filed for bankruptcy in the United States, raising concerns over the privacy of its collected DNA data and uncertainty over who would ultimately control and own that information (Kaur 2025).

The 23andMe data breach highlights the vulnerabilities of genomic data to cyberattacks, as a sensitive type of bioinformation that can reveal extensive information about humans, including their phenotype, health, and kinship, and as a type of data that is impossible to change if breached (Pulivarti et al. 2023). Such concerns about cybersecurity of genomic data form a key part of the growing discussions in academic and policy circles about cyber risks linked to the use of information, processes, and materials at the intersection of life sciences and digital technologies. As research and operations in the life sciences become increasingly digitized, and as governments invest heavily in artificial intelligence (AI) and machine learning to strengthen their position in the global bioeconomy, perceptions of cyber threats are becoming central to the construction of biological dangers. The COVID-19 pandemic has intensified such fears, due to a significant number of cyber espionage campaigns that targeted research organizations and companies working on vaccines and treatments, some of which were widely attributed to nation-state actors (Smith 2020).

This cyber-bio convergence, and the vulnerabilities associated with it, have given rise to the newly emerging concept and field of *cyberbiosecurity*. As introduced by scholarly literature in the life sciences, cyberbiosecurity, or bio-cybersecurity, is an “enterprise” or a “hybridised discipline” that addresses cyber risks and vulnerabilities engendered by digitizing biology (Murch et al. 2018). Such risks include, for example, cyberattacks that embed malware in DNA, corrupt gene sequencing, manipulate biomedical materials, or steal epidemiological data, potentially leading to the spread of diseases, the development of biological weapons, or even loss of life (Anjum et al. 2025). Although the trajectory of technological developments in the life sciences has been largely influenced by big technology corporations (Schuhmacher et al. 2022), research communities in these fields are the key drivers for the co-production of cyberbiosecurity as a field of research and policy analysis. This ongoing co-production of cyberbiosecurity hinges on two processes of legitimation: first, the conceptualization and enactment of “technical” knowledge and expertise through which actancy is defined; and second, the construction of a particular vision of a threatening “future” organized around a constel-

lation of security referent objects, with “life” itself at the center.

This article critically examines the sociotechnical imaginaries of cyberbiosecurity in life sciences scholarly literature since the inception of the concept in two biotechnology articles in 2017 and 2018 (Peccoud et al. 2017; Murch et al. 2018). As defined by Jasanoff, sociotechnical imaginaries refer to “collectively held, institutionally stabilized, and publicly performed visions of desirable futures, animated by shared understandings of forms of social life and social order attainable through, and supportive of, advances in science and technology” (Jasanoff 2015b, 4). While much of the literature on sociotechnical imaginaries in science and technology studies (STS) examines imaginaries of “desirable futures” that can only be attained through technological innovations and disruptive technologies (Csernatoni and Martins 2023), this article interrogates two competing yet interdependent imaginaries that underpin the emerging field of cyberbiosecurity. The first is a socio-technical imaginary that positions technology as a transformative and progressive force in the future of biology; the second is an (in)security imaginary that frames technology itself as an existential threat to life in a *threatened* and *threatening* future. Taken together, these imaginaries—and the intensifying cyber-bio convergence they co-produce—constitute an *informationalist challenge* to foundational questions in security research and policy analysis, which demands an intervention from international relations (IR) and security studies. Specifically, the article asks: How are competing and interdependent imaginaries of cyberbiosecurity constructed and mobilized in life sciences scholarship? And what implications do these imaginaries hold for core assumptions about actancy, referent objects, and uncertainty in security analysis?

The article proceeds in three sections. The first section examines the sociotechnical imaginaries surrounding digitization in the life sciences. It contextualizes the development of cyberbiosecurity as a field of research and policy analysis within two constructed inevitabilities that shape imaginaries of cyber-bio convergence: the inevitability of digitization as a positive force in the life sciences, and the inevitability of insecurities arising from that digitization. The second section examines the (in)security imaginaries of cyber-bio risks and their role as co-productive forces shaping the emerging field of cyberbiosecurity. It analyzes the epistemics of cyberbiosecurity and how actancy in the field is tied to understandings of “technical” expertise in knowledge production, which enables futuristic scenarios of cyber-bio dangers that prioritizes life as a referent object. The third section then moves to an analysis of the potential implications of cyber-bio imaginaries on the future of cybersecurity scholarship and governance. It argues that cyber-bio infrastructures introduce an “informationalist challenge” that reconfigures core assumptions about actancy, referent objects, and uncertainty, with significant ontological and epistemological implications for future research and policy across biosecurity, cybersecurity, and cyberbiosecurity.

Socio-Technical Imaginaries of Cyber-Bio Convergence

The growing link between imagination, innovation, and technology has driven an increased focus on the concept of socio-technical imaginaries in STS scholarship (Pfotenhauer and Jasanoff 2017). Socio-technical imaginaries refer to collectively shared and institutionally established visions of desirable futures, which are enacted and shaped by common understandings of the kinds of social life and order that scientific and technological progress can enable and sustain (

¹DTC-GT companies collect and analyze consumers' DNA samples to provide insights into their ancestry and health.

(Jasanoff 2015b, 4). These imaginaries are both products and instruments of co-production processes for science, technology, and society (Jasanoff 2015b). As such, the concept of sociotechnical imaginaries has been used to study various technology-related issues, including cybersecurity (Möllers 2021; Gjesvik and Szulecki 2023), cloud infrastructures (Baur 2023), platform security (Goede 2025), low-carbon warfare (Depledge et al. 2025), AI innovations (Chung 2025), biotechnology governance (Seay-Fleming 2021), and many others.

Sociotechnical imaginaries differ from “policy agendas” in that they are not explicit, issue-specific, or instrumentally designed. Rather than functioning as articulated programs of action, they operate at a deeper cultural level, residing in the background reservoir of norms, metaphors, and shared meanings from which actors derive and justify their policy preferences. Further, while they resemble “narratives” and “discourses” in identifying the boundaries of what is thinkable, sociotechnical imaginaries are distinguished by their entanglement with active exercises of power. They do not merely frame debates; they become materialized through the selection of policy priorities, the allocation of funding, the construction of infrastructures, etc. (Tidwell and Smith 2015).

Employing the lens of sociotechnical imaginaries makes it possible to examine the co-productive relationship between technologies and the future (Baur 2023). This lens provides a foundation for the analysis of the origins of many prevalent scientific ideas and technologies; how they are sustained through collectives that share understandings of the past, present, and future; and how such ideas become embedded in material practices and institutions (Jasanoff 2015c). Within this framework, imaginaries matter, not for how “accurate” they are as representations or forecasts of the future, but for what they reveal about how visions of the future are formed and contested, and for the potential performative effects these visions have on security governance (Depledge et al. 2025). Building on this body of literature, the sociotechnical imaginaries surrounding the transformative role of technologies in life sciences can be analyzed.

Digitization processes have been accelerated across various sectors, industries, and fields of research, with life sciences considered among the most significant. These processes have been driven by governments, researchers, entrepreneurs, and technology firms, who often wrap technological developments in grand narratives—e.g., utopian futures, community-building, and tech-based solutions to social problems—to advance particular visions of a technologized society (Mager and Katzenbach 2021). The ever-growing promises of technological innovation have long encouraged multiple stakeholders to invest in advancements in AI, robotics, and machine learning in order to digitize biology (Konrad and Böhle 2019; Richardson et al. 2019). In part, such investments have been fueled by the imaginary of “networked life” or “life as information,” in which biological systems can be analyzed, modified, and reprogrammed through information and communication technologies (ICTs) in increasingly sophisticated ways (Hester 2020). Further, these digitization processes reflect widespread belief in the fundamental role technologies play in enabling the scientific research critical to sustaining and advancing the bioeconomy, particularly biotechnology research and its diverse applications in manufacturing, agriculture, healthcare, and other domains (Philp 2021). The COVID-19 pandemic, for instance, demonstrated the importance of ICTs and data-driven approaches in times of global health emergencies, and how they became integral to screening and diagnosis of diseases, ge-

netic monitoring, and vaccine development (Benning et al. 2022).

Nowadays, many labs in the life sciences are being automated, transitioning into “smart labs,” with multiple networked devices that perform automatic functions and allow remote control and monitoring (Beutel and Lenk 2022). Technologies for data transmission and storages are also widely used to manage data streams generated by lab equipment, business information, financial data, research data, etc. (Reed and Dunaway 2019). This, in turn, has enabled what is referred to as “bio-foundries” or “cloud labs”: facilities in which AI-based software is utilized to automate the design and analysis of experiments, which is seen as essential to enhance productivity and reduce manufacturing costs and time (Elgabry and Johnson 2024). Several fields stand at the forefront of this cyber-bio convergence and this vision of digitized biology. For instance, synthetic biology is an important interdisciplinary field focusing on designing new artificial biological pathways, organisms, or devices, as well as modifying existing natural biological systems, thereby transforming food, materials, and medicine. Using ICTs to enable and facilitate research in gene editing, bioinformatics, and genome sequencing has been integral to developments in synthetic biology (Li et al. 2021). Now, the global synthetic biology market size is expected to reach \$148.93 billion by 2033, growing from an estimated \$20.01 billion in 2024 (Precedence Research 2024).

Importantly, imaginaries of technology’s transformative power in the life sciences have long been intertwined with *security imaginaries* around technological gaps among states, especially superpowers. The literature on security imaginaries in IR research, primarily developed by constructivist and poststructuralist approaches, predates the more recent turn toward sociotechnical imaginaries in STS literatures (Watts 2026). In some IR literatures, security imaginaries are defined as the shared memory, history, and cultural resources that actors draw upon when engaging with issues of foreign and security policy (Jacobsen and Nørgaard 2024). They are primarily a belief that threats and (in)security are social and political constructions that convey particular meanings, views, and identification of the “self” and the “other” (Pretorius 2008). However, this article draws on STS literatures that define security imaginary as a particular form of sociotechnical imaginary that foregrounds how technologies are developed and deployed under the rationale of security (Johnson 2024). Such literature analyzes how technology can be used as a solution for a security problem or framing technological progress as essential element of defense strategies (Winkler 2025). Here, it is important to note the role of the “other,” be it an adversary or a rival, and how it can enable the construction of shared collectives around particular imaginaries. For example, Klimburg-Witjes shows how the EU defense fund and initiatives for military-related research and innovations have been largely influenced by fears of technological gaps with the United States (Klimburg-Witjes 2023). In this context, the actual capabilities and limitations of technological systems matter less than these collectively imagined gaps and, subsequently, the desired vision of the future (Gerhold and Brandes 2021).

In fact, efforts to digitize biology are deeply linked to the global race for technological dominance notably in AI, machine learning, and their many associated domains, including advanced AI models, computing infrastructure, global technology standards, and the use of AI in physical systems like military equipment. In this competition, AI is seen as a winner-takes-all technology, promising huge economic, military, and geopolitical gains for the country that leads it

(Kahl 2026). This is clearly stated in the Trump administration's AI Action Plan entitled "Winning the AI Race," treating AI as a contest with only one possible winner, similar to the Cold War's logic in relation to space and nuclear power (Executive Office of the President of the USA 2025). This technological competition unfolds across many fields, with life sciences being among the most important. Over the past decade, countries such as the United States, China, Japan, and EU states have integrated biotechnology into their economic and security strategies through multiple policy documents (Nariyoshi Shinomiya and Tanaka 2025). Unsurprisingly, the global biotechnology market was valued at \$1.55 trillion in 2023 and is now projected to grow to \$2.44 trillion by 2028 and \$3.88 trillion by 2030 (Petit 2024). The United States and China, in particular, stand out due to their competitive approach to biotechnology, viewing it as a domain of both national security and, importantly, global *dominance*.

Since 2016, the United States has classified gene editing as a potential enabler of weapons of mass destruction, framing it as a major national security concern (Clapper 2016). In line with this, the National Security Commission on Emerging Biotechnology was established in 2022 to assess the implications of emerging biotechnologies for the Department of Defense (DoD). Several strategic documents—such as the 2022 National Security Strategy and the CHIPS and Science Act—explicitly position biotechnology, and especially competition with China, as central to US national security (117th US Congress 2022; The White House 2022). One major consequence was the China Initiative, launched by the US Department of Justice in 2018 to counter economic espionage by identifying suspected spies with ties to China in academia and industry. The initiative, which ran until 2022, disproportionately targeted Chinese-American scientists, resulting in the dismissal of around 250 researchers and legal actions against many others (Aloearhive and Guoarchive 2022). Similar fears of Chinese espionage surfaced in countries such as Canada and Australia, where the Australian Academy of Science condemned the trend as harmful to international collaboration (Hurst 2020; Austen 2024).

China has also approached digitizing biology as a domain of military significance since its 2013 Science of Military Strategy report (China Aerospace Studies Institute, Air University 2022). The 2020 edition further emphasized biotechnology as a "brand new territory for the expansion of national security," framing biology as part of the broader strategic competition among great powers (Tianliang et al. 2022). This reflects a strategy that positions technological research as a means of enhancing self-reliance and asserting sovereignty. In 2021, China introduced a comprehensive biosecurity law addressing threats such as the theft or misuse of biotechnologies, thereby reinforcing state control over genetics, biological resources, and related academic research. That same year, China's 14th Five-Year Plan for bio-economic development prioritized biosecurity risk prevention and control as a core objective (Biberman 2024). In parallel, the Chinese government has increased oversight and intervention in life sciences research. In 2018, for example, authorities seized exported genomic data and revoked licenses for two major international collaborations—one between UCLA and Shanghai Jiaotong University, and another between Oxford University and Peking University (Fouad 2024a). Additionally, China's strict data security laws, which extend to research data, have further constrained collaboration with foreign researchers (Matthews 2025).

Certain research industries have become key battlegrounds for these imaginaries of transformative life-science technologies, and for the perceived need to close "gaps" with

adversaries to secure the future. For example, in this US-China tech rivalry, the biopharmaceutical sector is a case in point. For years, Chinese biopharmaceutical firms depended on access to the United States for R&D collaborations, investment opportunities, and global market integration. In response, the Chinese government has significantly increased investment in domestic biopharma innovation to enhance self-reliance—an effort the United States now views as a national security concern. Notably, China already leads global production of active pharmaceutical ingredients for generic drugs (Gong 2025). To counter this growing influence, the United States has taken a series of strategic steps. In 2022, the Biden administration issued an executive order targeting Chinese biopharmaceutical companies. The United States also spearheaded the formation of the Bio-5 coalition—with the EU, India, Japan, and South Korea—to reduce reliance on Chinese inputs (The White House 2024). Now, the 2024 Biosecure Act prohibits entities receiving federal funding from using biotechnology supplied by companies linked to foreign adversaries, specifically China (118th Congress 2024).

Scholarly literature on cyberbiosecurity reinforces this long-established imaginary adopted by academic and policy communities of a better future enabled by the digitization of biology—one that is often justified through fears of adversaries seeking technological dominance as a security threat. Yet, unlike other technology-related domains where multiple counter-imaginaries critically challenge dominant imaginaries of digitization (e.g., see Kazansky and Milan 2021), the cyberbiosecurity imaginary does not contest narratives of digitizing biology. Instead, cyberbiosecurity actively builds upon and reinforces existing imaginaries of digitizing biology to construct the security imaginaries of cyber-bio risks. For instance, one study argues that "the field of Cyberbiosecurity is an example of the emerging concept of bioconvergence" (Greenbaum 2023, 2). Bioconvergence here is defined as "the idea that there are many opportunities for profitable synergy between the biosciences and other areas of technology, hi-tech in particular, that were formerly wholly independent of the biosciences" (Greenbaum 2023, 2).

That is, instead of opposing digitization processes, cyberbiosecurity imaginaries build on two inevitabilities: the inevitability of digitization as a positive force in the life sciences, and the inevitability of insecurities resulting from this digitization. Put differently, technology in cyberbiosecurity imaginaries is understood as both a *security enabler* and a *security threat*. It is a security enabler particularly when digitizing biology is imagined as a "race" with national security consequences that adversaries should not be allowed to win. It is simultaneously a security threat because the more digitized a system is, the more vulnerable it becomes to cyberattacks, and the more serious the consequences of those attacks would be when they take place. As argued by Joly, the politics of new and emerging technologies is often understood as a politics of potential and possibility—one oriented toward the future and shaped by struggles over legitimacy in which both optimistic promises and warnings of catastrophe play central roles (Joly 2015). This argument can be extended to cyberbiosecurity as well, although here the warnings of catastrophe are tied directly to technology itself. The next section examines these (in)security imaginaries of cyber-bio risks and their role as co-productive forces shaping the emerging field of cyberbiosecurity.

(In)Security Imaginaries of Cyber-Bio Risks

Thinking about cybersecurity in the life sciences is not a new phenomenon; it goes back to 2014, when discussions

around safeguarding the increasingly digitized bioeconomy started to emerge. During that time, a working group was funded by the US Federal Bureau of Investigation's (FBI) Weapons of Mass Destruction Directorate, in collaboration with the American Association for the Advancement of Science (AAAS) and the UN Interregional Crime and the Justice Research Institute (UNICRI). The aim of the group was to analyze the security risks associated with big data in the life sciences, in conversation with multiple academics and industry partners, including Microsoft, IBM, Google, and Amazon (AAAS-FBI-UNICRI 2014). Thereafter, an ad hoc committee appointed by the National Academies of Sciences, Engineering, and Medicine organized multiple workshops on the implications of technology vulnerabilities on the bioeconomy, at the request of the FBI (NASEM 2015). Various incidents of previous data breaches were presented in these workshops as catalysts for more sophisticated future attacks. A key example is the data breach impacting the health insurance provider Anthem Inc.—now known as Elevance Health—in 2015, compromising personally identifiable information and medical records of 78.8 million individuals (Riley 2015). Yet, these were scattered efforts that did not necessarily culminate in a clear imaginary of a new cyberbiosecurity landscape that requires peculiar tools and policies to address.

Understanding the co-production process of cyberbiosecurity as an (in)security imaginary requires an examination of the role of the specific groups of researchers who coined the term, developed funding applications to advance it, and engaged in marketing efforts for the concept/field to gain political and scientific recognition. As Hilgartner argues, although multiple future sociotechnical imaginaries can exist simultaneously, the ones that tend to persist are usually those advanced by what might be described as “sociotechnical vanguards” (Hilgartner 2015). Sociotechnical vanguards are small groups of actors that act together in an intentional manner to push for a particular vision of the future to be accepted, be it in global or local spheres, even in the existence of dialectal relationships among contradictory imaginaries (Hilgartner 2015; Jasanoff 2015b). These groups need to mobilize significant resources, whether technological or institutional, for imaginaries to be adopted and shared in collective decision-making (Depledge et al. 2025).

This framework is useful in analyzing the origins of cyberbiosecurity as developed by socio-technical vanguards in research communities in life sciences, and the multiple efforts to solidify the imaginaries of cyber-bio risks it presents. Randal Murch, a former academic at Virginia Tech, who is one of the key founders of the field/concept of cyberbiosecurity, traces the origins of cyberbiosecurity in a dedicated book chapter published in 2023 (Murch 2023). Murch explains that the concept was first developed by him and two other Virginia Tech researchers, who recognized that conventional cybersecurity could not adequately protect digitally dependent biological operations. The three researchers began to develop the concept, working toward “further communications and marketing seeking visibility and funding” (Murch 2023, 7). The concept gained momentum through engagement with academic, commercial, and US government stakeholders, culminating in a federally funded systems analysis of a bioprocessing facility that helped formalize the discipline.

With funding from the US DoD, a team of life sciences academics from three universities in the United States started a project to specifically study the security of critical information in the life sciences, to provide recommendations on biodefense to the DoD, and ultimately, to develop the foundations of cyberbiosecurity as a new field (University

of Nebraska-Lincoln 2017). Two biotechnology articles were published thereafter in 2017 and 2018 to officially coin the term “cyberbiosecurity” as “an emerging new discipline” (Peccoud et al. 2017; Murch et al. 2018). Other foundational steps to the field included the first cyberbiosecurity workshop in 2017 and an e-book that attracted global attention, both led by the same group of researchers, and involving multiple government and industry stakeholders (Murch 2023). A significant number of academic publications appeared thereafter, which demonstrates the growing importance of cyberbiosecurity in academic discussions (Potter and Palmer 2023).

This growing academic interest is echoed in policy circles too, where cyberbiosecurity is starting to be recognized as a critical security concern. The US government remains at the forefront of these discussions, spearheading multiple initiatives that reflect the emergence of a military-academic-industrial complex co-producing cyberbiosecurity. For instance, in 2019, the Bipartisan Commission on Biodefense, composed of former senior government officials, convened a panel to address “cyber-bio convergence” with experts from academia, policy, and industry (Bipartisan Commission on Biodefense 2019). More recently, the National Cybersecurity Center of Excellence and the National Institute of Standards and Technology launched a project and issued guidance focused on cybersecurity for genomic data (Pulivarti et al. 2023). A series of summits also bring together multiple stakeholders, such as the 2023 summit at the Georgia Tech Research Institute (Guise et al. 2024), and the 2025 summit hosted by the Johns Hopkins University Applied Physics Laboratory in collaboration with the Bioeconomy Information Sharing and Analysis Center (BIO-ISAC).² The BIO-ISAC is an international non-profit organization that brings together industry, academia, and government to improve biotech cybersecurity, with partners like Google Cloud, MIT, and Lincoln Laboratory.

Similar policy discussions are starting to gain prominence beyond the United States too. For example, the United Kingdom's recent Biological Security Strategy mentioned the convergence of biosciences with robotics, machine learning, and AI as a catalyst for “cyberbiosecurity risks” (UK Government 2023). This aligns with the establishment of the ministerially chaired UK Biosecurity Leadership Council, gathering academics, industry experts from life sciences and biotechnology, and government officials to discuss multiple policy priorities, including cyberbiosecurity (Department for Science, Innovation and Technology 2023). On a multilateral organizational level, the European Union Agency for Cybersecurity (ENISA) is another important example, following its identification of cyberbiosecurity as an “urgent” issue in 2022 (European Union Agency for Cybersecurity 2022). ENISA published a report that includes “cyberbiosecurity” as one of four selected case studies for AI security risks in biology, and conducted a roundtable discussion on cyberbiosecurity to identify key research gaps (European Union Agency for Cybersecurity 2023, 2024).

Actancy and the Epistemics of Cyberbiosecurity

In complex, multidisciplinary fields, defining actancy—i.e., who has the capacity to act and influence security co-production, is not an easy task. In recent years, many literatures in critical cybersecurity research began to approach cybersecurity as an “assemblage” that comprises a com-

²Official website of the Cyberbiosecurity Summit 2025: <https://www.cyberbiosecuritysummit.org>.

plex configuration of cybersecurity actors beyond the state (Collier 2018; Stevens 2020). In such assemblages, a complex and constantly shifting network of actors and security communities establishes what constitutes “knowledge” and “truth” about incidents, which consequently shape security practices (Jacobsen 2020; Egloff and Dunn Cavelty 2021). In cyberbiosecurity imaginaries, actancy is closely tied to a specific epistemic authority assumed by the sociotechnical vanguards of this burgeoning field. As argued by STS research on the “genomic revolution” and the transformative scientific and social change it brought, sociotechnical vanguards often adopt a “visionary role,” presenting themselves as having superior insights into emerging developments and how to employ them for desirable futures. In doing so, they shape expectations, evoke hopes and fears, and help bring particular futures into existence (Hilgartner 2017).

Similarly, the socio technical vanguards of cyberbiosecurity in research communities introduced the field as fundamentally distinct from biosecurity and cybersecurity, requiring peculiar knowledge and expertise at the intersection of life sciences, biosecurity, and technical cybersecurity (Murch et al. 2018). While biosecurity focuses on preventing the spread of pathogens and harmful organisms that cause infectious diseases, cyberbiosecurity reframes these threats by focusing specifically on those arising from the integration of biological (animate) and cyber (inanimate) systems (Dixon 2021). Murch argues that the introduction of cyberbiosecurity as a new field/concept is important because existing biosecurity efforts have been too narrowly focused on biological rather than digital threats, and that no government agency has seriously addressed the vulnerabilities created by increasingly interconnected cyber-bio infrastructures (Murch 2023).

Here, cyberbiosecurity imaginaries rely primarily on a particular conceptualization and enactment of “technical” knowledge and expertise in establishing actancy. In several respects, this understanding parallels security discourses that have shaped both cybersecurity and biosecurity. Given its complex technological nature and the fast pace of its evolution, knowledge required to understand cybersecurity is not readily available for everyone, including IR and security scholars. This has granted legitimacy and epistemic authority to certain fields of expertise and research, such as computer science and computer engineering, to construct “technical” and “expert” imaginaries around cybersecurity based on close engagement with cyber operational environments (Hansen and Nissenbaum 2009; Cristiano et al. 2024). Similarly, the imaginaries of global health security and biosecurity have long promoted a form of governance anchored in scientific knowledge, in which scientific expertise is seen to bind together an imagined global community. Because the authority of science is framed as purely epistemic rather than political in many such imaginaries, the political work through which scientific sovereignty is achieved becomes obscured, appearing instead as a natural and inevitable extension of expert knowledge (Enticott and Wilkinson 2013; Hurlbut 2017).

In cyberbiosecurity literature, claims of epistemic authority and technical expertise are evident in the way the field has been framed as “peculiar,” “new,” and distinct from existing cybersecurity and biosecurity discussions. Proponents contend that current knowledge about vulnerabilities, threats, and protective measures in cybersecurity and biosecurity is insufficient for securing cyber-bio infrastructures, which they describe as presenting a “unique problem” (Schabacker et al. 2019). As one study puts it, biotechnology sectors remain constrained by security perspectives rooted in biosecu-

rity and biosafety, while cybersecurity specialists often lack familiarity with biotechnology (Mueller 2021). Other studies examining cyberbiosecurity risk perceptions within the biotechnology sector similarly conclude that cyber-bio risks are neither well known nor well understood among biotechnology and cybersecurity experts, calling for more coordinated efforts to develop this emerging field (Millett et al. 2019).

Accordingly, cyberbiosecurity imaginaries add an extra level of technicality to what has been widely understood as inherently technical fields—i.e., biosecurity and cybersecurity—by framing the kind of threats the two fields address, methods they employ, and knowledge they possess as primarily “traditional” (DiEuliis 2020). Ultimately, this call for new forms of knowledge and expertise gives an epistemic authority to small groups of life scientists with particular technical knowledge around cyber-bio risks to assume the role of security actors. As put by Murch, the group of experts who engaged in fund-raising activities to introduce the concept of cyberbiosecurity, and who succeeded in securing the US DoD funding, are now being called upon on various occasions by multiple stakeholders to represent the field and enhance its feasibility (Murch 2023). Yet, unlike traditional sociotechnical vanguards in other fields who present themselves as experts possessing superior insight into the desirable potentials of technologies (Hilgartner 2015), cyberbiosecurity vanguards claim expertise in recognizing the *undesirable* future security implications of technology itself—even as they continue to advocate for ongoing digitization.

Security Referent Objects in a Threatening and Threatened Future

The question of what constitutes security referent objects—i.e., who/what is threatened and therefore requires protection—has always been a central question in security analysis and one that has significant implications on security policies and practices (Zhou 2022). Within cyberbiosecurity imaginaries, these referent objects can be broadly divided into two categories. The first includes entities that have already been targeted by cyberattacks in the past or present, and which serve to legitimize representations of cyberbiosecurity threats. The second encompasses hypothetical future scenarios in which “life” per se is threatened, where imagined threats are mobilized to justify pre-emptive security measures.

Imaginaries of inevitable future cyber-bio risks are fundamentally shaped by particular perceptions of an insecure past and present. As Pfotenhauer and Jasanoff argue, sociotechnical imaginaries of desirable futures must be grounded in an undesirable present or in a diagnosed deficit that is collectively and publicly felt (Pfotenhauer and Jasanoff 2017). In this view, the past and future exist in a “complex dialectic” within sociotechnical imaginaries, with the past serving as a “prologue” and a “site of memory” that is mobilized and reinterpreted to make sense of the present and to shape the future (Jasanoff 2015b). Yet, the undesirable past and present invoked in the cyberbiosecurity literature do not arise from a context in which technologies promise to remedy a collectively experienced “deficit”; rather, the deficit in cyber-bio imaginaries is an environment of “insecurity” generated by the technologies themselves.

The insecurity environment imagined in cyberbiosecurity literature is represented in the rising number of cyber incidents worldwide, some of which have affected hospitals, research institutions, and pharmaceutical companies as key referent objects. These previous cyber incidents are used by

many literature to build an imaginary of how the future may look like if no sufficient interventions are implemented to prevent them (Chatzis et al. 2023). A key example of a widely referenced cyber incident in this regard is the NotPetya ransomware in 2017 (Reed and Dunaway 2019; Crawford et al. 2023; Anjum et al. 2025). Although it was allegedly launched by the Russian government to disrupt government institutions in Ukraine, NotPetya self-propagated and affected a large number of entities around the world, among which was the pharmaceutical giant Merck & Co. The ransomware brought down 40,000 of the company's machines, with an estimated loss of \$1.4 billion (Frost 2024). Likewise, the WannaCry ransomware in 2017 significantly impacted the United Kingdom's National Health Service, leading to wide-scale operational disruptions (Abed et al. 2024). Both incidents were catalysts for discussions on the importance and complexities of cybersecurity in healthcare.

Arguably, the greatest influence on cyberbiosecurity imaginaries was the COVID-19 pandemic. During the pandemic, many hospitals, pandemic response institutions, and health care providers were targeted by cyberattacks. For example, one of the most serious attacks targeted Ireland's Health Service Executive, in which a ransomware encrypted around 80 percent of system data, thus disrupting the national diagnostic imaging platform and halting radiotherapy services in five major centers (BBC News 2021). Further, cyber espionage campaigns targeting pharmaceutical companies and research institutions developing vaccines and treatments raised serious concerns about the security of digitised bioinformation. Nation-state actors, notably China, Russia, North Korea, and Iran, have been accused of launching these espionage campaigns against healthcare and research institutions across the United States, United Kingdom, Canada, France, India, and South Korea (Burt 2020). While these cyber incidents did not cripple vaccine development or significantly impact pandemic management, they highlighted the strategic value of bioinformation, the risks of its misuse in cyber operations, and the intricate links between geopolitics, cybersecurity, and global health (Fouad 2024a).

However, at the heart of envisioned potential cyber-bio threats lies "life" itself as a referent object. Life could be threatened, for example, if potential cyberattacks lead to developing biological weapons, spreading diseases, or causing death (Chatzis et al. 2023; Crawford et al. 2023; Kambouris et al. 2025). Since such attacks have not yet occurred, their ongoing construction remains hypothetical, often supported by experiments conducted to demonstrate the potential for these attacks taking place in the future. In fact, perceptions of cyberattacks leading to loss of life in medical fields are not entirely new and have been traditionally tied to the "internet of medical things" (Pycroft and Aziz 2018). A well-known example is the US Vice President Dick Cheney having his heart defibrillator's wireless function disabled in 2007 for fear it could be compromised leading to a heart attack (BBC News 2013). More recently, experimental studies have been demonstrating how cyberattacks targeting, for example, the algorithms used for diagnosing skin cancer from images could distort the diagnosis, even if altering just a few pixels in biopsy images (Jogani et al. 2022). Another experiment showed how attacks using deep-learning techniques could interfere with medical scans to tamper with or remove evidence of medical conditions (Mirsky et al. 2019).

What distinguishes cyberbiosecurity imaginaries, nonetheless, is the perception of potential cyber-bio attacks combining both animate (biological) and inanimate (cyber) substrates (Dixon 2021). A key example is what is referred to as "biological malicious software," or "bio-malware"

(Elgabry and Johnson 2024). Since DNA is now computerized, i.e., being read, sequenced, written, and synthesized using ICTs, a study showed that physical DNA itself can now become a vector to compromise bioinformatics programs (Ney 2019). This could allow the remote access of a system, without the scientists' awareness (Elgabry 2023). Among the most widely cited experiments in this regard is one that was conducted in 2017 in which scientists injected malware in DNA strands and managed to subsequently control the computer system used to analyze the DNA (Regalado 2017). Another experimental study showed the possibility of obtaining leaked information from DNA synthesizer through acoustic side channels by using microphones positioned near the device—a method referred to as the "Oligo-snoop" attack (Faezi et al. 2019). Some studies construct scenarios of a terrorist or a sophisticated malicious actor designing a malware that could potentially interfere with the process of generating synthetic DNA, thus producing dangerous substances (Farbiash and Puzis 2020). This is linked to what one study calls "cyber-biocrime," viewed as a "future biotechnology crime" that may combine both computers and the internet together with biological or biochemical materials (Elgabry et al. 2020).

All these experiments co-produce an imaginary of a future in which cyber-bio attacks are a matter of "when" not "if" (Kambouris et al. 2025). Accordingly, cyberbiosecurity is being constructed as "pre-emptive security," aiming at preventing an imminent cyber-bio attack from taking place. The pre-emptive security measures advocated by proponents in the field range from standard cybersecurity practices, e.g., strong passwords, multifactor authentication, etc., to calls for extensive government intervention (Fouad 2024a). Multiple voices demand intervention from governments, national security agencies, and militaries in securing life sciences infrastructure and imposing more security measures on conducting research (DiEuliis 2023). This is based on a widespread assumption that scientists and private sector partners have "a surprising level of naiveness" and are incapable of securing such systems on their own (Peccoud et al. 2017). For example, some studies propose incorporating cyberbiosecurity in military's and intelligence agencies' cybersecurity strategies as a "national security issue" (George 2019); establishing "super-governmental framework for academic research" that specifies guidelines for data sharing in the life sciences (Potter and Palmer 2023); treating bioinformation databases as "global critical infrastructure" (Caswell et al. 2019); among others. All such calls for more government involvement go in line with how the field evolved in the first place and the military and governmental funding that made it possible.

The Informationalist Challenge

Cyberbiosecurity imaginaries have successfully attracted policy attention and sparked important debates about future cyberbiosecurity risks and the measures needed to prevent or mitigate them, as previously discussed. Through the technical expertise of the cyberbiosecurity sociotechnical vanguards and future-oriented imaginaries of cyberbio risks, the field succeeded in securing government attention, attracting funding, and migrating the concept from scientific discussions into broader policy circles. Yet, beyond immediate policy solutions, a key problem in prevailing imaginaries of cyberbiosecurity—and in how the topic is discussed outside IR, security studies, and the critical scholarship in cybersecurity and biosecurity—is that they obscure crucial ontological and epistemological questions about the informational nature of both fields. This article argues that cyber-

biosecurity imaginaries and their associated cyber-bio convergence generate an *informationalist challenge* that future research and policy in biosecurity, cybersecurity, and cyber-biosecurity must engage with. Addressing this challenge has broader implications for security analysis, particularly regarding how we conceptualize the security referent object, forms of actancy, and the nature and management of uncertainty.

For many years, security scholarship across both cybersecurity and biosecurity has been grappling with the so-called “material turn” in IR, which questions the long-standing assumption that agency is the exclusive domain of human actors. Various philosophical paradigms have evolved to challenge the binary distinction between human subjects and non-human objects; a shift often referred to as “the material turn,” “non-human turn,” “thing studies,” or “new materialism.” These paradigms include object-oriented ontology (Bryant 2011; Bogost 2012; Harman 2018), vital materialism (Bennett 2009), agential realism (Barad 2003), and actor-network theory (Latour 2005; Law and Singleton 2005). While these frameworks differ in how they conceptualize the relationship between humans and non-humans, they share a critical perspective on the Anthropocene’s dominant status in the study of social and political phenomena (McDonald and Mitchell 2017; Kaltofen 2018). They theorize matter as an active, politically significant force with meaning beyond social, political, or economic structures. This perspective blurs the traditional human/non-human divide, necessitating a closer study of materiality (Braidotti 2013; Salter 2015).

Several studies on biosecurity adopt this new materialist or more-than-human approach to the analysis of biosecurity as an inherently multi-system governance in which the unpredictable mobilities of nonhumans, be them diseases, viruses, pests, etc., shape security analysis (Barry 2024). They show, for example, how microbes and pathogens engage in “bordering practices,” by dictating safe and unsafe spaces, beyond the control of humans or states bordering policies (du Plessis 2017). Given the entanglement of human health with other species, some studies call for new forms of global health governance that embrace this ontological plurality (Voelkner 2022). Beyond pathogens and diseases, some studies focus on other forms of “matter,” or non-material things, which have performative influences on global health security and biosecurity, including, for example, protective clothing of health workers (Krause 2021; Wezel 2024), emergency operations centers (Elbe 2023), among others. In cybersecurity too, there is a growing body of research that engages with this “materialist challenge” (Markussen 2025). Many such literatures approach cybersecurity as an assemblage of complex configuration of actors and alliances, be they conceptual, material, social, or textual (Stevens 2020; Egloff and Dunn Cavelty 2021). Some literatures specifically focus on the role of non-human agency, by conceptualizing malware as “political actors” co-shaping cybersecurity politics and practices (Balzacq and Dunn Cavelty 2016; Dwyer 2021; Liebetau and Christensen 2021; Fouad 2022).

The increasing cyber-bio convergence, co-produced by the aforementioned imaginaries, introduces another important ontological and epistemological challenge to the future of security analysis: an *informationalist challenge*. As argued by Stevens, new materialist approaches have not adequately integrated “information” as a category within their conceptual frameworks, and thus have not engaged with key philosophical debates about the ontology of information and its relationship to the “matter” theorized by new materialism (Stevens 2012). Building upon insights from the philosophy of information, philosophy of computation, and philoso-

phy of biology, cyber-bio infrastructures can be conceptualized as complex informational infrastructures. Such informational nature challenges the modes of constructing cyber-biosecurity as a distinct field and necessitates more critical engagement with the informational challenges that such infrastructures co-produce to both scholarship and policy.

On one side, information lies at the heart of the philosophy of computation and all the *technologies, sciences, and practices* constitutive of the “cyber” (Fouad 2025a). Here, information can be defined as syntactic information (i.e., signals, codes, bits, etc.); semantic information (i.e., ideas or meaning embedded in bits or signals); and pragmatic information (i.e., when those ideas are novel to the recipient) (Deacon 2010). At a fundamental level, the various sciences and technologies that make up what we call “cyberspace” are all grounded in information. Computer science, for instance, is frequently described as “the body of knowledge of information-transforming processes” or even more simply as “the study of information” (Primiero 2016). The view of computing as a “science of information” dates to the 1960s, when the field was characterized as “the systematic study of the ontologies and epistemology of information structures” (Primiero 2016, 104). Computers themselves are typically understood as machines that store and process information, whether represented in bits (0 s and 1 s), switch positions (on and off), or electrical voltages (high and low) (Dunn 2008). At the physical level, computers rely on binary states—ones and zeros, or electrical and electromagnetic signals—to function, which is often interpreted as structural *information*. Beyond this, operational and instructional forms of information link the hardware to specific tasks through programming languages and user interfaces that mediate interactions between humans and machines. In short, every aspect of how computing devices work involves information, whether in the form of bits, code, programming languages, algorithms, or even the intentions guiding a computer engineer (Fresco and Wolf 2016).

On the other side, information plays a key theoretical and foundational role in biological systems beyond the use of metaphors/analogies such as “codes,” “transcriptions,” and “translations” (Barbieri 2003; Levy 2011). Many biologists argue that genes exert their causal influence by virtue of carrying information about the products they help generate. In biology, information is often seen as central to understanding life. Whole-organism traits, including complex behaviors, are commonly described as being specified or encoded by genetic information. Likewise, many cellular processes—and even the full developmental trajectory of an organism—are framed as the execution of a genetic “program.” In evolutionary theory, genes are sometimes viewed as entities that are, in a meaningful sense, made of information (Godfrey-Smith 2007). Developmental biology can be understood as examining how genetic information is transformed into the structure of the adult organism, while evolutionary biology investigates how that information originated and changed over time. DNA is often said to hold information shaped by natural selection, information that specifies the amino-acid sequences of proteins. In a broader way, both DNA and proteins are also seen as carrying instructions—or a developmental program—for building the organism. Natural selection acting on organisms then modifies the information contained in the genome (Smith 2000). Although this view of the centrality of information in biology is opposed by some biologists and philosophers (Griffiths 2001; Boniolo 2003), the argument in relation to cyber-bio infrastructures being inherently informational appears more straightforward.

That is, cyber-bio infrastructures could be conceptualized as informational infrastructures with complex configurations of human and non-human agency. Despite the fact that malware is human-made, it reflects many of the same challenges to human control as viral pathogens—extending beyond mere pathological comparisons to biological categorization (Dwyer 2024). For example, self-propagating malware can spread autonomously across information systems, beyond the intentionality of their human creators, often causing unpredictable and unintended consequences (Fouad 2022). Thus, cyber-bio infrastructures problematize the category of the “biological” and the “digital” as well as what constitutes a “virogenic activity.” Computerized DNA, and subsequent fears of encoding malware in synthesized strands of DNA in bio-malware, as previously explained, is a prominent example of the increasing blurring lines between the biological/physical and the digital/virtual into a broader informational framework (Elgabry and Johnson 2024). As argued by Winickoff, “Modern biotechnology puts the very idea of nature into question. Its techniques hybridize agent and object, the given and made, the lively and inert” (Winickoff 2015). Cyber systems extend this hybridity by adding the “digital” to the mix, embedding it within a broader informational framework.

Information is deeply entangled with complexity and self-organization (Fouad 2024b). Complexity itself is difficult to define, but it is broadly understood across sciences as a property of systems that are composed of multiple interdependent parts interacting in non-linear ways (Burgin and Calude 2016). This non-linearity makes the outcomes of such systems unpredictable: even minor changes can trigger disproportionately large effects (Merali 2006). Many complex systems also display behavior at the boundary between order and disorder, where interactions produce novel and unexpected outcomes; what is often referred to as emergence (Baranger 2000). Emergence results from the dynamic interactions among components of these complex systems, and cannot be deduced from analyzing parts in isolation (Bousquet and Curtis 2011). In many ways, emergence is treated as an ontological feature of self-organizing systems; one that is real and impactful, regardless of whether humans recognize it (Morçöl 2013). Because emergent behavior arises autonomously, it challenges assumptions that environments can be fully known, analyzed, or controlled by humans.

Cyber-bio infrastructures that integrate digital networks, biological processes, and automated decision-making exhibit the same complex, self-organizing properties found in advanced information systems. They consist of many interdependent components (e.g., digital sensors, genomic databases, AI analytics, biological agents) whose interactions generate non-linear and often unpredictable outcomes. These infrastructures can autonomously adapt to new data, reconfigure themselves in response to environmental change, and produce emergent behaviors that cannot be inferred from any single part. The parallels between biological and digital substrates deepen this complexity: DNA functions as a biochemical code capable of mutation, recombination, and replication, much like software that can rewrite itself, propagate errors, or interact with other code in unanticipated ways. In both domains, the “programs” governing system behavior do not operate under the full oversight of any single human actor. Instead, system-level effects emerge from countless local interactions occurring across laboratories, databases, algorithms, and living organisms.

In this context, cyberbiosecurity imaginaries can be critiqued for their limited engagement with the ontological and epistemological questions raised by this informational com-

plexity. By framing cyber-bio risks primarily as a technical or policy problem, these imaginaries sideline deeper inquiries into how security can be conceptualized in relation to increasingly complex informational systems. Cyber-bio infrastructures, understood as informational infrastructures, generate contingencies and configurations of (im)materiality that challenge assumptions of human control over security environments. In doing so, they destabilize the pre-emptive security logics and claims of existential threat that cyberbiosecurity imaginaries often advance. This calls for a re-imagination of security analysis that considers the ontological uncertainties produced by such complex informational systems, and that interrogates what these uncertainties mean for security actancy and agency and for long-standing assumptions about human control embedded in pre-emptive or preventive security logics. Future cybersecurity, biosecurity, and cyberbiosecurity research must, therefore, grapple with several key questions: How do we conceptualize actancy and agency in ways that account for these complex informational infrastructures? How do ontological uncertainties and emergence reshape our understanding of security and defense logics? And what becomes of claims of existentiality in security analysis when the referent object is the (im)material “informational”?

Conclusion

Although cyberbiosecurity is growing in prominence as novel security issue in academic and policy debates, there is currently limited engagement from IR, security studies, or social sciences at large with this burgeoning field. The article addresses this gap by analyzing the peculiarities of the sociotechnical imaginaries of cyberbiosecurity in the life sciences literature since its emergence in 2017, highlighting the more nuanced and often conflicted views of technology they present. While much of the STS literature on sociotechnical imaginaries examines narratives that frame technology as a positive force for future progress and transformation, this article showed how cyberbiosecurity is co-produced through two contradictory yet mutually reinforcing imaginaries: the inevitability of digitizing biology and the inevitability of insecurity engendered by that digitization per se. The first imaginary advances a sociotechnical vision of digital technologies as progressive and transformative to the future of biological innovation, and essential in maintaining technological dominance over rivals. Many fields stand at the forefront of such digitization, including synthetic biology, genomics, laboratory automation, etc. The second constructs an (in)security narrative in which those very technologies generate existential vulnerabilities; blurring the boundaries between the *digital* and the *biological*, the *virtual* and the *physical*, and between *animate* and *inanimate* substrates.

In this sense, the (in)security imaginaries surrounding cyberbio risks differ from existing analyses of security imaginaries around technology in the literature, which often portray technology as a tool for addressing security problems or preventing technological gaps with rivals that could pose national security threats. Perceived insecurities in cyberbiosecurity go beyond conceptualizations of self/other and allies/enemies, neither are they limited to fears of falling behind if adversaries acquire particular technologies. Rather, cyberbiosecurity imaginaries depict future threats produced, in part, by digitization itself, in the form of cybersecurity vulnerabilities that could be exploited through cyberattacks targeting cyber-bio infrastructures. The more digitized a system is, the more vulnerable it becomes to cyberattacks, and the more serious the consequences of those attacks would be

when they take place. Put differently, technology in and of itself becomes both a security enabler and a security threat.

Together, these imaginaries produce visions of an imminent threatening and threatened future in which cyberattacks may spread disease, manipulate biological systems, or cause physical harm. Accordingly, cyberbiosecurity is framed as a preventative and pre-emptive security project, premised on future-oriented imaginaries of security risks that are predicted to materialize at the intersection of cyber and biological systems. By drawing on past and present cyber incidents, alongside experimental demonstrations of hypothetical cyber-bio risks, cyberbiosecurity scholars establish a sense of urgency and certainty concerning future insecurities. These future-oriented imaginaries have been crucial in securing government attention, attracting funding, and migrating the field from scientific discussions into broader policy circles. In this process, an influential group of researchers in the life sciences acted as the sociotechnical vanguard of cyberbiosecurity, consolidating epistemic authority in shaping how the “cyber-bio” future is imagined.

However, cyberbiosecurity raises important ontological and epistemological challenges that extend beyond the technical limits of these imaginaries. This article argues that cyber-bio infrastructures operate as complex informational systems that destabilize categories central to security analysis, including agency, uncertainty, and the referent object of security. As the boundaries between life and information and between material and digital processes become increasingly entangled, future research and policy in cybersecurity and biosecurity must engage with the ontological and epistemological implications of this convergence. Addressing this informational challenge will require the development of new analytical tools capable of explaining how uncertainty is co-produced, how (non-)human actancy is distributed, and how security itself must be rethought in the context of cyber-bio convergence. That is, the article calls for a more reflexive engagement from IR and security studies with the futures imagined in and through cyberbiosecurity that are being actively constructed through the interplay of sociotechnical and (in)security imaginaries.

Acknowledgment

I am grateful to the three anonymous reviewers and the editors of *Global Studies Quarterly* for their thoughtful and constructive comments, which significantly contributed to improving this article.

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